UGV-UAV Coordination: Takeoff, Landing and Formation Control

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***Abstract*—This article presents a formation control paradigm between an Unmanned Ground Vehicle (UGV) and an Unmanned Aerial Vehicle (UAV). The objective for the UGV is to track the desired trajectory and the UAV coordinates with the UGV for autonomous takeoff and landing. The aspects of landing of the UAV as well as taking off from the moving UGV are explored. The stability of the proposed controllers for both UGV and UAV is verified through the Lyapunov stability criterion. The effectiveness of the proposed controllers are shown through extensive simulation results including a circular, a sinusoidal, and an eight shaped trajectories. The comparative study also shows the effectiveness of the proposed work.**

***Index Terms*—Unmanned Ground Vehicle, Unmanned Aerial Vehicle, Heterogeneous System, Trajectory Tracking, Formation Control, Dynamic Modelling**

I. Introduction

Controlling of an Unmanned Aerial Vehicle (UAV) has become an important aspect of research in the recent time. An UAV is useful in many applications like surveillance, map building, delivery in emergency situations (food, medicines in natural catastrophe), or delivery of e-commerce items. Now, there are several constraints which restrict the UAV to manueveur for long time like battery capacity or fuel. So, to recharge the battery or refuelling, the UAV needs to land and it has to take off from that ground station. The ground station could be stationary or moving. The ground station could be an Unmanned Ground Vehicle (UGV) also. So, this problem needs special attention due to its complexity and wide scope of applications.

A distributed formation controller was proposed in [1]. The authors developed two controllers for the UGV and the UAV respectively. The UAV maintained a desired formation with the UGV while moving on a desired trajectory. It showed that the UAV was able to land on and takes off from the UGV. A sliding mode controller based guidance of an UAV for landing and taking off on a stationary and moving ground vehicle was described in [2]. A vision based formation controller design of a heterogeneous system was given in [3]. In [4], the authors developed a time-varying formation controller based on the dynamic modeling of the UGV and the UAVs. The formation controller of a heterogeneous system consisting of the UGV and the UAV was also proposed in [5]. A coordinated air-ground vehicle was described in [6]. In [7], the authors proposed a formation protocol design for a heterogeneous system. The agents were communicated through directed graph and in leader-follower approach. Similar to that, another formation controller was proposed in [8] for a heterogeneous system where the network topology was a weighted directed graph. In [9], a distributed time-varying formation controller was proposed for a class of heterogeneous linear multi-agent systems. The multi-agent system contained a virtual leader, multi leaders, and followers. The simulation results validated the effectiveness of the proposed distributed time-varying controller. The authors in [10] developed an observer based adaptive time-varying formation controller for a heterogeneous system. The proposed observer was able to estimate the state matrix of the leader agent in finite time. In article [11], consensus tracking of a group of heterogeneous agents was presented. To achieve the objective, an interdependent model of the heterogeneous system was built. A formation controller was proposed in [12] for a class of nonholonomic robots with docking capability. The robots were capable of dynamic formation to avoid the collision with obstacles.

This paper contributes on these following aspects

1) A formation controller is proposed for both the UGV and the UAV.
2) The UAV generates the control input based on information regarding the state of the UGV.
3) The simulation results show that the proposed controller is effective and guides the UAV for successful taking off and landing from/on the UGV.
4) The effectiveness of the proposed controller is validated through a comparative study.

The rest of article is organized as follows: the modelling of both the UGV and the UAV is described in section II, followed by the design of controllers for both the systems in section III and IV respectively. Section V refers to the simulation results of the proposed controller. Section VI discusses about the conclusion and possible scope of extensions of the proposed work.

II. Modelling of the heterogeneous system

The modelling of both the UGV and the UAV is presented here.

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A. Modelling of an Unmanned Ground Vehicle

This section describes the model of the UGV system. The UGV parameters are taken from [13]. The linear and angular velocities of the UGV are denoted as u_g and ω_g. u_{ref} and ω_{ref} are the control inputs to the system. The model of the UGV is presented through equation (1) [13], [14].

$$\begin{bmatrix}\dot{x}_g\\ \dot{y}_g\\ \dot{\psi}_g\\ \dot{u}_g\\ \dot{\omega}_g\end{bmatrix}=\begin{bmatrix}u_g\cos\psi_g-a\omega_g\sin\psi_g\\ u_g\sin\psi_g+a\omega_g\cos\psi_g\\ \omega_g\\ \frac{\theta_{3_g}}{\theta_{1_g}}\omega_g^2-\frac{\theta_{4_g}}{\theta_{1_g}}u_g\\ -\frac{\theta_{5_g}}{\theta_{2_g}}u_g\omega_g-\frac{\theta_{6_g}}{\theta_{2_g}}\omega_g\end{bmatrix}+\begin{bmatrix}0&0\\0&0\\0&0\\ \frac{1}{\theta_{1_g}}&0\\ 0&\frac{1}{\theta_{2_g}}\end{bmatrix}\begin{bmatrix}u_{ref}\\ \omega_{ref}\end{bmatrix} \tag{1}$$

x_g and y_g represent the position of the UGV. ψ_g is the heading angle of the UGV. θ_g is a vector, represented as $\theta_g=[\theta_{1_g}\ \theta_{2_g}\ \theta_{3_g}\ \theta_{4_g}\ \theta_{5_g}\ \theta_{6_g}]^T$ [13]. a is the distance between the centre of two wheels and point of interest of the UGV [13].

B. Modelling of an Unmanned Aerial Vehicle

The positional and the attitude subsystems can be represented separately as $p=[x,y,z]^T$ and $\vartheta=[\phi,\theta,\psi]^T$ with respect to the earth frame and body frame respectively. ϕ, θ and ψ are the Euler angles representing the roll, pitch, and yaw angles of the UAV. The velocity and acceleration of the UAV are denoted as $\dot{p}=[\dot{x},\dot{y},\dot{z}]^T$ and $\ddot{p}=[\ddot{x},\ddot{y},\ddot{z}]^T$ respectively. The angular velocity and acceleration of the UAV with respect to the body frame are denoted as $\dot{\vartheta}=[\dot{\phi},\dot{\theta},\dot{\psi}]^T$ and $\ddot{\vartheta}=[\ddot{\phi},\ddot{\theta},\ddot{\psi}]^T$ respectively. The UAV model and the parameter values are taken from [15]. The model of the UAV is described in equation (2) [15].

$$\begin{cases}\ddot{x}=U_x\frac{U_1}{m}\\ \ddot{y}=U_y\frac{U_1}{m}\\ \ddot{z}=\frac{C_\phi C_\theta}{m}U_1-g\\ \ddot{\phi}=\dot{\theta}\dot{\psi}\left(\frac{I_y-I_z}{I_x}\right)-\frac{J_r}{I_x}\dot{\theta}\omega+\frac{l}{I_x}U_2\\ \ddot{\theta}=\dot{\phi}\dot{\psi}\left(\frac{I_z-I_x}{I_y}\right)+\frac{J_r}{I_y}\dot{\phi}\omega+\frac{l}{I_y}U_3\\ \ddot{\psi}=\dot{\phi}\dot{\theta}\left(\frac{I_x-I_y}{I_z}\right)+\frac{1}{I_z}U_4\end{cases} \tag{2}$$

where, $U_x = (C_\phi S_\theta C_\psi+S_\phi S_\psi)$ and $U_y = (C_\phi S_\theta S_\psi-S_\phi C_\psi)$, here $S_{(.)}=\sin(.)$ and $C_{(.)}=\cos(.)$. The values of the roll, pitch, and yaw angle are from $\phi\in(-\frac{\pi}{2},\frac{\pi}{2})$, $\theta\in(-\frac{\pi}{2},\frac{\pi}{2})$, and $\psi\in(-\pi,\pi)$ respectively and $\omega=\omega_2+\omega_4-\omega_1-\omega_3$ [15]. $\omega_1,\omega_2,\omega_3,\omega_4$ are the speed of the rotors. I_x,I_y and I_z are the moment or inertia along the respective axes. m is the mass of the UAV, and l is the distance between center of the UAV to the center of the rotor. g stands for the gravitational acceleration. J_r represents the total moment of inertia along the propeller axis. U_1,U_2,U_3, and U_4 are the control inputs along z-axis, and roll, pitch, yaw angle respectively.

III. Controller for the UGV

This section describes the steps to design the tracking controller for the UGV and formation controller for the UAV. The model of the UGV described in equation (1) can be divided into its kinematic and dynamic model. The design of both kinematic and dynamic controller is given in the next section.

A. Kinematic controller

The kinematic model of the UGV system is described in equation (3).

$$\begin{bmatrix}\dot{x}_g\\ \dot{y}_g\\ \dot{\psi}_g\end{bmatrix}=\begin{bmatrix}\cos\psi_g & -a\sin\psi_g\\ \sin\psi_g & a\cos\psi_g\\ 0 & 1\end{bmatrix}\begin{bmatrix}u_g\\ \omega_g\end{bmatrix} \tag{3}$$

Now considering the linear motion of the UGV, it can be expressed as $h=[x_g\ y_g]^T$. Hence,

$$\dot{h}=\begin{bmatrix}\dot{x}_g\\ \dot{y}_g\end{bmatrix}=\begin{bmatrix}\cos\psi_g & -a\sin\psi_g\\ \sin\psi_g & a\cos\psi_g\end{bmatrix}\begin{bmatrix}u_g\\ \omega_g\end{bmatrix} \tag{4}$$

The inverse kinematics of the UGV is given by,

$$\begin{bmatrix}u_g\\ \omega_g\end{bmatrix}=\begin{bmatrix}\cos\psi_g & \sin\psi_g\\ -\frac{1}{a}\sin\psi_g & \frac{1}{a}\cos\psi_g\end{bmatrix}\begin{bmatrix}\dot{x}_g\\ \dot{y}_g\end{bmatrix} \tag{5}$$

The kinematic control laws for the linear (u^c_{ref}) and the rotation (ω^c_{ref}) velocities are expressed as

$$\begin{bmatrix}u^c_{ref}\\ \omega^c_{ref}\end{bmatrix}=\begin{bmatrix}\cos\psi_g & \sin\psi_g\\ -\frac{1}{a}\sin\psi_g & \frac{1}{a}\cos\psi_g\end{bmatrix}\begin{bmatrix}x_{d_{2_g}}+k_{x_g}e_{x_g}\\ y_{d_{2_g}}+k_{y_g}e_{y_g}\end{bmatrix} \tag{6}$$

$x_{d_{2_g}}$ and $y_{d_{2_g}}$ are the desired velocities of the UGV along the x and y axes respectively. k_{x_g} and k_{y_g} are the positive constant gains. Similarly, e_{x_g} and e_{y_g} are positional errors of the UGV along x and y axes respectively. The objective of designing this controller is to generate a reference control input for the dynamic controller.

Stability analysis: The stability of the proposed controller is ensured using the Lyapunov stability criteria. In order to minimize the velocity errors, this condition must be followed

$$\begin{bmatrix}u^c_{ref}\\ \omega^c_{ref}\end{bmatrix}=\begin{bmatrix}u_g\\ \omega_g\end{bmatrix} \tag{7}$$

So,

$$\begin{bmatrix}\dot{e}_{x_g}\\ \dot{e}_{y_g}\end{bmatrix}+\begin{bmatrix}k_{x_g}e_{x_g}\\ k_{y_g}e_{y_g}\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix} \tag{8}$$

where, $\dot{e}_{x_g}$ and $\dot{e}_{y_g}$ are the velocity errors along the respective axes of the UGV. Now considering $e_h=[e_{x_g}\ e_{y_g}]^T$. Equation (8) can be rewritten as

$$\dot{e}_h=\begin{bmatrix}\dot{e}_{x_g}\\ \dot{e}_{y_g}\end{bmatrix}=-\begin{bmatrix}k_{x_g}e_{x_g}\\ k_{y_g}e_{y_g}\end{bmatrix} \tag{9}$$

Now defining a Lyapunov function as

$$V_1=\frac{1}{2}e_h^Te_h \tag{10}$$

The time derivative of equation (10) can be written as

$$\dot{V}_1=-k_{x_g}(e_{x_g})^2-k_{y_g}(e_{y_g})^2 \tag{11}$$

while k_{x_g} and k_{y_g} are positive constant values then $\dot{V}_1\le 0$. So it can be concluded that, the proposed controller is stable.

B. Dynamic controller

In this section, the design of controller considering the dynamic model of the UGV is explained. The dynamic model of the UGV can be expressed as

$$\begin{bmatrix}\dot{u}_g\\ \dot{\omega}_g\end{bmatrix} = \begin{bmatrix}\frac{\theta_{3_g}}{\theta_{1_g}}\omega_g^2 - \frac{\theta_{4_g}}{\theta_{1_g}}u_g\\ -\frac{\theta_{5_g}}{\theta_{2_g}}u_g\omega_g - \frac{\theta_{6_g}}{\theta_{2_g}}\omega_g\end{bmatrix} + \begin{bmatrix}\frac{1}{\theta_{1_g}} & 0\\ 0 & \frac{1}{\theta_{2_g}}\end{bmatrix}\begin{bmatrix}u_{ref}\\ \omega_{ref}\end{bmatrix} \tag{12}$$

Defining the the error along linear velocity e_{u_g} as

$$e_{u_g} = u^c_{ref} - u_g \tag{13}$$

The time derivative of the equation (13) yields,

$$\dot{e}_{u_g} = \dot{u}^c_{ref} - \dot{u}_g \tag{14}$$

Now, the controller along linear motion u_{ref} is designed as

$$u_{ref} = \theta_{1_g}\left(\frac{\theta_{4_g}}{\theta_{1_g}}u_g - \frac{\theta_{3_g}}{\theta_{1_g}}\omega_g^2 + \dot{u}^c_{ref} + k_u sgn(e_{u_g})\right) \tag{15}$$

Stability analysis: This section provides the necessary proof of stability of the controller which is designed for stable operation of the UGV. Considering a Lyapunov function V_2 as

$$V_2 = \frac{1}{2}(e_{u_g})^2 \tag{16}$$

The time derivative of equation (16) yields,

$$\dot{V}_2 = e_{u_g}\left(\dot{u}^c_{ref} - \frac{\theta_{3_g}}{\theta_{1_g}}\omega_g^2 + \frac{\theta_{4_g}}{\theta_{1_g}}u_g - \frac{u_{ref}}{\theta_{1_g}}\right) \tag{17}$$

Now replacing the value of u_{ref} from equation (15) in equation (17) it becomes,

$$\dot{V}_2 = -k_u|e_{u_g}| \tag{18}$$

while $k_u > 0$, then $\dot{V}_2 \leq 0$. Hence, the proposed controller of the UGV is asymptotically stable. Following this similar approach, the stability of the proposed controller can be validated for the UGV along the angular motion. The angular motion controller ω_{ref} can be designed as

$$\omega_{ref} = \theta_{2_g}\left(\frac{\theta_{5_g}}{\theta_{2_g}}u_g\omega_g + \frac{\theta_{6_g}}{\theta_{2_g}}\omega_g + \dot{\omega}^c_{ref} + k_\omega sgn(e_{\omega_g})\right) \tag{19}$$

IV. CONTROLLER FOR THE UAV

This section describes, the steps to design the controllers for both the positional and the attitude subsystems for the UAV to track the desired trajectory and taking off and landing from/on the UGV.

A. Controller for the positional subsystem

The design of controller along positional subsystem is described here for the z-axis. A similar steps need to be taken to design the controllers along the x and y axes. Lets, $z_1 = z$ and $z_2 = \dot{z}_1$, so the dynamics of the UAV along the z axis can be rewritten as follows

$$\ddot{z}_1 = \dot{z}_2 = (\cos\phi\cos\theta)\frac{U_1}{m} - g \tag{20}$$

Now, $\dot{z}_1$ and $\ddot{z}_1$ are velocity and acceleration along the z-axis respectively of the UAV. The positional and velocity tracking errors e_{z_1} and e_{z_2} are defined as

$$e_{z_1} = z_1 - z_{d_1} \tag{21}$$

$$e_{z_2} = z_2 - z_{d_2} \tag{22}$$

where, z_{d_1} represents the desired position of the UAV and z_{d_2} is the virtual control, which is expressed as

$$z_{d_2} = \dot{z}_{d_1} - k_{z_1}e_{z_1} \tag{23}$$

k_{z_1} is a positive constant. Now taking the time derivative of equation (21), yields,

$$\dot{e}_{z_1} = z_2 - \dot{z}_{d_1} \tag{24}$$

The derivative of the velocity error e_{z_2} with respect to time yields,

$$\dot{e}_{z_2} = \dot{z}_2 - \dot{z}_{d_2} \tag{25}$$

Now, defining the functions e_{d_z} and $\hat{e}_z$ as

$$e_{d_z} = |z_g - z_1| + e_{z_1} - \delta_z \tag{26}$$

$$\hat{e}_z = e_{z_2} + e_{d_z} \tag{27}$$

z_g is the position of the UGV along the z-axis. δ_z is the desired distance between the UGV and UAV along the z-axis. The Lyapunov based control input U_1 is designed as

$$U_1 = \frac{m}{c_\phi c_\theta}\left(\dot{z}_{d_2} - \dot{e}_{d_z} + g - k_{z_2}sgn(\hat{e}_z)\right) \tag{28}$$

The stability of the control input is explained in the next section.

Stability analysis of the positional subsystem: The stability of the proposed positional controller along z-axis is ensured by the Lyapunov stability criteria. Now, a Lyapunov function is defined as

$$V_z = \frac{1}{2}\left(\hat{e}_z\right)^2 \tag{29}$$

The time derivative of equation (29) gives,

$$\dot{V}_z = \hat{e}_z\left((\cos\phi\cos\theta)\frac{U_1}{m} - g - \dot{z}_{d_2} + \dot{e}_{d_z}\right) \tag{30}$$

Substituting the value of U_1 in equation (30), yields

$$\dot{V}_z = -k_{z_2}\left|\hat{e}_z\right| \tag{31}$$

while, k_{z_2} is a positive constant, so $\dot{V}_z \leq 0$. Hence, it can be concluded that the controller along the z-axis is stable. The controllers along the x and the y axes can also be designed following the similar approach. The control inputs U_x and U_y are given as

$$U_x = \frac{m}{U_1}\left(\dot{x}_{d_2} - \dot{e}_{d_x} - k_{x_2}sgn(\hat{e}_x)\right) \tag{32}$$

$$U_y = \frac{m}{U_1}\left(\dot{y}_{d_2} - \dot{e}_{d_y} - k_{y_2}sgn(\hat{e}_y)\right) \tag{33}$$

B. Desired Euler angles

The desired roll and pitch angles are described in equation (34), and (35) respectively [16]. The desired yaw angle is $\psi_{d_1} = 0$ for the UAV.

$$\phi_{d_1} = sin^{-1}(U_{e_x} \sin\psi - U_{e_y} \cos\psi) \tag{34}$$

$$\theta_{d_1} = sin^{-1}\left(\frac{U_{e_x}}{\cos\phi\cos\psi} - \frac{\sin\psi\sin\phi}{\cos\phi\cos\psi}\right) \tag{35}$$

where, U_{e_x} and U_{e_y} are defined as

$$U_{e_x} = \frac{(\dot{x}_{d_1} - \dot{x}_1)m}{U_1}, \quad U_{e_y} = \frac{(\dot{y}_{d_1} - \dot{y}_1)m}{U_1} \tag{36}$$

C. Controller for the attitude subsystem

In this section, the design of controller along the attitude subsystem is described. The steps to design the controller along the roll angle (ϕ) is explained elaborately. To design the controllers along the pitch (θ) and the yaw angles (ψ), similar approach needs to be followed. Now consider, $\phi_1 = \phi$ and $\dot{\phi_1} = \phi_2$. Then the dynamics of the roll angle can be written in the form as

$$\ddot{\phi}_1 = \dot{\phi}_2 = \dot{\theta}\dot{\psi}\left(\frac{I_y - I_z}{I_x}\right) - \frac{J_r}{I_x}\dot{\theta}\omega + \frac{U_2}{I_x}l \tag{37}$$

The angular velocity and acceleration of the UAV along roll angle is represented as $\dot{\phi}_1$ and $\ddot{\phi}_1$ respectively. Now defining the angular and velocity error along the roll angle e_{ϕ_1} and e_{ϕ_1} as

$$e_{\phi_1} = \phi_1 - \phi_{d_1} \tag{38}$$

$$e_{\phi_2} = \phi_2 - \phi_{d_2} \tag{39}$$

ϕ_{d_1} is the desired roll angle of the UAV and ϕ_{d_2} is the virtual control. It is expressed as

$$\phi_{d_2} = \dot{\phi}_{d_1} - k_{\phi_1} e_{\phi_1} \tag{40}$$

k_{ϕ_1} is a positive constant. Now the time derivative of roll angle error is expressed as

$$\dot{e}_{\phi_1} = e_{\phi_2} - k_{\phi_1} e_{\phi_1} \tag{41}$$

Time derivative of e_{ϕ_2} is

$$\dot{e}_{\phi_2} = \dot{\phi}_2 - \dot{\phi}_{d_2} \tag{42}$$

The control input U_2 is designed based on the Lyapunov approach for a stable operation as

$$U_2 = \frac{I_x}{l}\left(-e_{\phi_1} + \dot{\phi}_{d_2} - \dot{\theta}\dot{\psi}\frac{I_y - I_z}{I_x} + \frac{J_r}{I_x}\dot{\theta}\omega - k_{\phi_2} sgn(e_{\phi_2})\right) \tag{43}$$

Stability analysis of the attitude subsystem: The stability of the proposed attitude controller along the roll angle is validated through the Lyapunov stability criteria. The Lyapunov function candidate is considered as

$$V_\phi = \frac{1}{2}\left(e_{\phi_1}\right)^2 + \frac{1}{2}\left(e_{\phi_2}\right)^2 \tag{44}$$

The differential form of equation (44) is

$$\begin{aligned} \dot{V}_\phi = &-k_{\phi_1} e_{\phi_1}^2 + e_{\phi_2}\left(\dot{\theta}\dot{\psi}\left(\frac{I_y - I_z}{I_x}\right) + \frac{U_2}{I_x}l\right) \\ &+ e_{\phi_2}\left(e_{\phi_1} - \frac{J_r}{I_x}\dot{\theta}\omega - \dot{\phi}_{d_2}\right) \end{aligned} \tag{45}$$

Substituting the value of U_2 from equation (43), yields

$$\dot{V}_\phi = -k_{\phi_1} e_{\phi_1}^2 - k_{\phi_2}\left|e_{\phi_2}\right| \tag{46}$$

while, k_{ϕ_1} and k_{ϕ_2} are positive constants, so $\dot{V}_\phi \leq 0$. Hence, it can be concluded that the proposed controller along the roll angle is stable. Following the similar approach, the controller for the pitch angle (θ) and the yaw angle (ψ), U_3 and U_4 can also be developed respectively as

$$U_3 = \frac{I_y}{l}\left(-e_{\theta_1} + \dot{\theta}_{d_2} - \dot{\phi}\dot{\psi}\frac{I_z - I_x}{I_y} - \frac{J_r}{I_y}\dot{\phi}\omega - k_{\theta_2} sgn(e_{\theta_2})\right) \tag{47}$$

$$U_4 = I_z\left(-e_{\psi_1} + \dot{\psi}_{d_2} - \dot{\phi}\dot{\theta}\tfrac{I_x - I_y}{I_z} - k_{\psi_2} sgn(e_{\psi_2})\right) \tag{48}$$

V. SIMULATION RESULTS

The simulation results of the formation control between the UGV and the UAV are presented here. The proposed formation controller is tested on three situations, a circular shaped, a sinusoidal shaped, and an eight shaped desired trajectories. The parameters to simulate the proposed formation controller are taken as $k_{z_1} = k_{z_2} = k_{x_1} = k_{y_1} = k_{\phi_1} = k_{\theta_1} = k_{\psi_1} = 2, k_{\phi_2} = k_{\theta_2} = k_{\psi_2} = 0.001$, $k_u = k_\omega = 10, k_{x_g} = k_{y_g} = 15$. $k_{x_2} = 5, k_{y_2} = 2$ for the circular and the sinusoidal shaped trajectories, and $k_{x_2} = k_{y_2} = 5$ for the eight shaped trajectory. The value of the parameters of the UGV and the UAV are given in Table I. The initial values of the UGV and the UAV for all the situations are given in Table II. σ is an offset and the value of σ is taken as 0.2 meters.

A. A circular shaped trajectory tracking

In this section, both the UGV and the UAV are tracking circular shaped desired trajectories. The autonomous landing, takeoff, and formation control aspects are shown in this experiment. Initially, the UAV starts from a certain height and it comes down to reach the desired trajectory in formation with the UGV. The desired trajectory for the UAV is placed at a height of 2 meters. After 17 seconds, the UAV initiates the landing and landed on the UGV and stays there upto 27 seconds. After 27 seconds, the UAV takes off from the UGV and tracks the desired circular shaped trajectory for the rest of the time. The trajectories tracked by both the UGV and the UAV are shown in Fig. 1. The positional errors along respective axes are shown in Fig. 2. It signifies that the UAV successfully has landed on the UGV and took-off from the UGV and tracked the desired trajectory.

B. A sinusoidal shaped trajectory tracking

In this application, the landing of the UAV on the UGV is explored while both of them are tracking sinusoidal shaped desired trajectories. Similar to the previous experiment, here also the UAV starts from a certain height and comes down to

TABLE I
UAV AND UGV PARAMETER VALUES [14], [15]

Parameters	$m(kg)$	$l(m)$	$g(m/s^2)$	$I_x(kg.m^2)$	$I_y(kg.m^2)$	$I_z(kg.m^2)$	θ_{1_g}	θ_{2_g}	θ_{3_g}	θ_{4_g}	θ_{5_g}	θ_{6_g}	a
Values	0.5	0.2	9.81	4.85×10^{-3}	4.85×10^{-3}	8.81×10^{-3}	0.26038	0.25095	-0.00049969	0.99646	0.002629	1.0768	0.5

TABLE II
INFORMATION REGARDING DESIRED TRAJECTORIES

Desired trajectory (m)		Initial values	
		Positions	angle
Circular shaped	$x_{g_d} = 10\sin t,\ x_{d_1} = x_{g_d}$ $y_{g_d} = 10\cos t,\ y_{d_1} = y_{g_d}$ $z_{g_d} = 0,\ z_{d_1} = z_{g_d} + 2, (0 < t \le t_1)$ $z_{d_1} = z_{g_d} + \sigma, (t_1 < t \le t_2)$ $z_{d_1} = z_{g_d} + 2, (t_2 < t \le t_3)$	$x_g = 1,\ x_1 = 5$ $y_g = 9,\ y_1 = 4$ $z_g = 0,\ z_1 = 4$	$\phi = 0$ $\theta = 0$ $\psi = 0$
Sinusoidal shaped	$x_{g_d} = t,\ x_{d_1} = x_{g_d}$ $y_{g_d} = 2\sin(t/3),\ y_{d_1} = y_{g_d}$ $z_{g_d} = 0,\ z_{d_1} = z_{g_d} + 2, (0 < t \le t_1)$ $z_{d_1} = z_{g_d} + \sigma, (t_1 < t \le t_2)$	$x_g = 1,\ x_1 = 8$ $y_g = 0,\ y_1 = 4$ $z_g = 0,\ z_1 = 5$	$\phi = 0$ $\theta = 0$ $\psi = 0$
Eight shaped	$x_{g_d} = 10\cos t\sin t,\ x_{d_1} = x_{g_d}$ $y_{g_d} = 10\sin t,\ y_{d_1} = y_{g_d}$ $z_{g_d} = 0,\ z_{d_1} = z_{g_d} + 2$	$x_g = 0,\ x_1 = 2$ $y_g = 0,\ y_1 = 8$ $z_g = 0,\ z_1 = 0$	$\phi = 0$ $\theta = 0$ $\psi = 0$

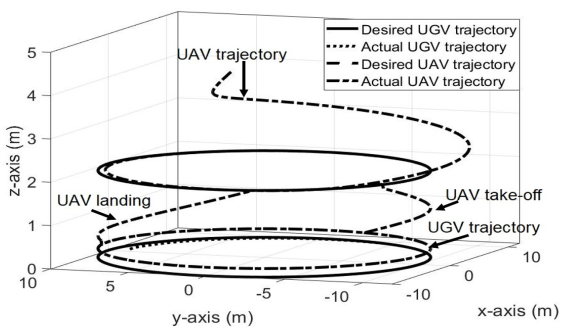

Fig. 1. A circular shaped trajectory tracking by the UGV and landing and takeoff of the UAV on/from the UGV while maintaining the desired formation

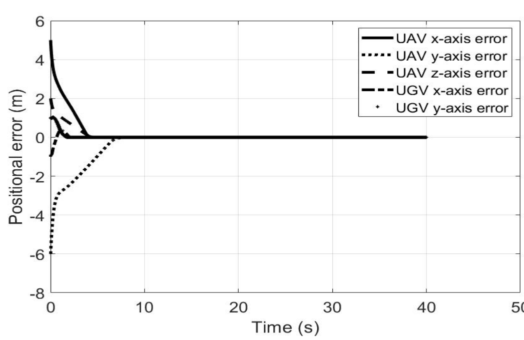

Fig. 2. Positional errors of the UGV and the UAV

track the desired trajectory at $z_{d_1} = 2$ meters. The UAV stays on the desired trajectory upto 20 seconds, and starts landing after that. Once the UAV landed on the UGV, it stays over it, for the rest of the time. Fig. 3 illustrates the desired and the actual trajectories the UGV and the UAV along with the safe landing of the UAV on the UGV. The positional errors of the UGV and UAV along the respective axes are shown in Fig. 4. As they are starting from random positions so the initial positional errors are comparatively high, though the

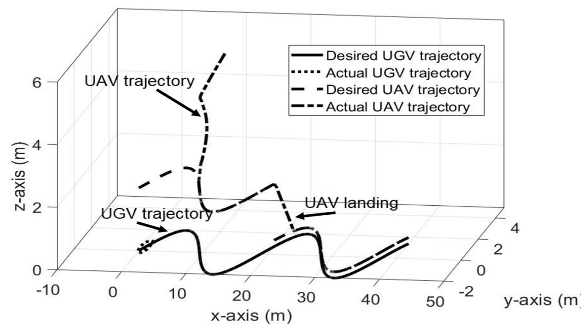

Fig. 3. A sinusoidal trajectory tracking by the UGV and landing of the UAV on the UGV while maintaining the desired formation

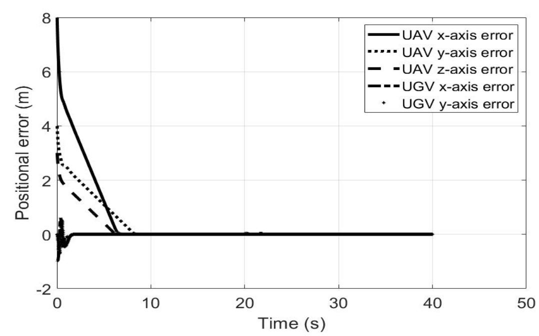

Fig. 4. Positional errors of the UGV and the UAV

errors eventually converge towards zero using the proposed formation controller.

C. An eight shaped trajectory tracking

In this section, the UGV and the UAV are tracking eight shaped desired trajectories. Fig. 5 shows that the UGV and the UAV have successfully tracked the desired trajectories in formation. The positional errors of both the UGV and the UAV converge towards zero as shown in Fig. 6. It signifies that, both the UGV and the UAV tracked their respective desired trajectories while maintaining the desired formation.

D. Comparative studies

The effectiveness of the proposed trajectory tracking controller for the UGV system is validated through the comparative study with the article [17]. In [17], the authors proposed a modified Nonlinear Model Predictive Controller (NMPC) and compared the results with an Adaptive Dynamic Controller (ADC), classic Nonlinear Model Predictive Controller (NMPC). Here, the objective for the UGV is to track a circular shaped and an eight shaped desired trajectories. The initial position of the UGV and the reference trajectories along with the parameter values are taken from [17]. The detailed comparative results are given in the Table III and IV.

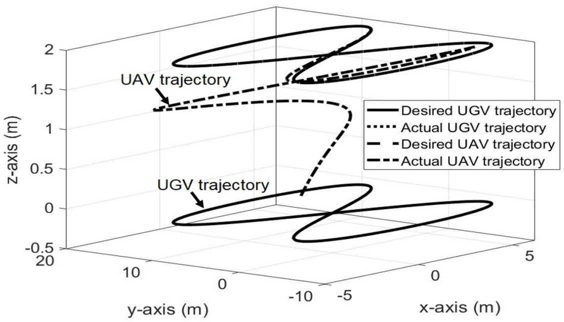

Fig. 5. An eight shaped trajectory tracking by the UGV and the UAV while maintaining the desired formation

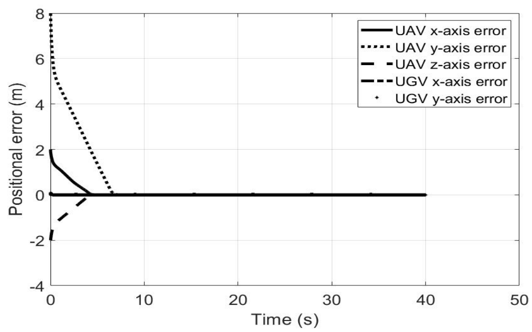

Fig. 6. Positional errors of the UGV and the UAV

TABLE III
COMPARATIVE RESULTS FOR A CIRCULAR SHAPED TRAJECTORY

Controller	IAE	ISE	ITSE
ADC	20.99	4.65	125.11
NMPC	17.58	3.89	108.78
Controller proposed in [17]	6.37	1.26	12.08
Proposed controller	1.1540	0.3796	0.3237

The proposed controller is evaluated in terms of the Integral Absolute Error (IAE), Integral Square Error (ISE), and Integral Time Square Error (ITSE). It is seen in the Table III and IV that there are significant improvements in the results.

VI. CONCLUSION

This paper presents a formation controller for the heterogeneous system which consists of an UGV and an UAV. Controllers are presented for both of them. The UGV is tracking the desired trajectory and the UAV is maintaining a desired formation with the UGV. It is also depicted that the UAV is able to takeoff and land on the UGV. The comparative results show that the proposed controller works satisfactorily. In future, this work can be extended for multiple UGVs and UAVs. The effect of unknown disturbances on system dynamics will also be explored.

TABLE IV
COMPARATIVE RESULTS FOR AN EIGHT SHAPED TRAJECTORY

Controller	IAE	ISE	ITSE
ADC	11.23	0.99	47.30
NMPC	28.99	5.34	533.19
Controller proposed in [17]	5.11	0.21	17.19
Proposed controller	4.4654	0.6841	8.5278

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